CSE2023 NON-GRADED ASSIGNMENT #1

Solve the following questions from course book. (Discrete Mathematics and Discrete Mathematics and Its Applications, 7th Ed. by Kenneth Rosen)

1- Let *p* and *q* be the propositions

p: You drive over 70 miles per hour.

q: You get a speeding ticket.

Write these propositions using *p* and *q* and logical connectives (including negations).

a) You do not drive over 70 miles per hour.
b) You drive over 70 miles per hour, but you do not get a speeding ticket.
c) You will get a speeding ticket if you drive over 70 miles per hour.
d) If you do not drive over 70 miles per hour, then you will not get a speeding ticket.
e) Driving over 70 miles per hour is sufficient for getting a speeding ticket.
f) You get a speeding ticket, but you do not drive over 70 miles per hour.
g) Whenever you get a speeding ticket, you are driving over 70 miles per hour.

2- For each of these sentences, state what the sentence means if the logical connective or is an inclusive or (that is, a disjunction) versus an exclusive or. Which of these meanings of or do you think is intended?

a) To take discrete mathematics, you must have taken calculus or a course in computer science.
b) When you buy a new car from Acme Motor Company, you get $2000 back in cash or a 2% car loan.
c) Dinner for two includes two items from column A or three items from column B.
d) School is closed if more than 2 feet of snow falls or if the wind chill is below −100.

3- You are eligible to be President of the U.S.A. only if you are at least 35 years old, were born in the U.S.A, or at the time of your birth both of your parents were citizens, and you have lived at least 14 years in the country. Express your answer in terms of e: “You are eligible to be President of the U.S.A.,” a: “You are at least 35 years old,” b: “You were born in the U.S.A,” p: “At the time of your birth, both of your parents where citizens,” and r: “You have lived at least 14 years in the U.S.A.”

4- Are these system specifications consistent? “Whenever the system software is being upgraded, users cannot access the file system. If users can access the file

system, then they can save new files. If users cannot save new files, then the system software is not being upgraded."

5- Show that $(p \rightarrow q) \vee (p \rightarrow r)$ and $p \rightarrow (q \vee r)$ are logically equivalent.

6- Show that $(p \wedge q) \rightarrow r$ and $(p \rightarrow r) \wedge (q \rightarrow r)$ are not logically equivalent.

7- Let Q(x) be the statement "x + 1 > 2x." If the domain consists of all integers, what are these truth values?

a) $Q(0)$ **b)** $Q(-1)$ **c)** $Q(1)$
d) $\exists x Q(x)$ **e)** $\forall x Q(x)$ **f)** $\exists x \neg Q(x)$
g) $\forall x \neg Q(x)$

8- Suppose that the domain of the propositional function $P(x)$ consists of −5, −3, −1, 1, 3, and 5. Express these statements without using quantifiers, instead using only negations, disjunctions, and conjunctions.

a) $\exists x P(x)$ **b)** $\forall x P(x)$
c) $\forall x((x \neq 1) \rightarrow P(x))$
d) $\exists x((x \geq 0) \wedge P(x))$
e) $\exists x(\neg P(x)) \wedge \forall x((x < 0) \rightarrow P(x))$

9- Express each of these statements using quantifiers. Then form the negation of the statement, so that no negation is to the left of a quantifier. Next, express the negation in simple English. (Do not simply use the phrase "It is not the case that.")

a) Some old dogs can learn new tricks.
b) No rabbit knows calculus.
c) Every bird can fly.
d) There is no dog that can talk.
e) There is no one in this class who knows French and Russian.

10- Show that these statements are inconsistent: "If Sergei takes the job offer then he will get a signing bonus." "If Sergei takes the job offer, then he will receive a higher salary." "If Sergei gets a signing bonus, then he will not receive a higher salary." "Sergei takes the job offer."

11- Let *L(x, y)* be the statement "x loves y," where the domain for both x and y consists of all people in the world. Use quantifiers to express each of these statements.

a) Everybody loves Jerry.
b) Everybody loves somebody.
c) There is somebody whom everybody loves.
d) Nobody loves everybody.

e) There is somebody whom Lydia does not love.
f) There is somebody whom no one loves.
g) There is exactly one person whom everybody loves.
h) There are exactly two people whom Lynn loves.
i) Everyone loves himself or herself.
j) There is someone who loves no one besides himself or herself

12- Determine the truth value of each of these statements if the domain of each variable consists of all real numbers.

a) $\forall x \exists y(x^2 = y)$ **b)** $\forall x \exists y(x = y^2)$
c) $\exists x \forall y(xy = 0)$ **d)** $\exists x \exists y(x + y \neq y + x)$
e) $\forall x(x \neq 0 \rightarrow \exists y(xy = 1))$
f) $\exists x \forall y(y \neq 0 \rightarrow xy = 1)$
g) $\forall x \exists y(x + y = 1)$
h) $\exists x \exists y(x + 2y = 2 \wedge 2x + 4y = 5)$
i) $\forall x \exists y(x + y = 2 \wedge 2x - y = 1)$
j) $\forall x \forall y \exists z(z = (x + y)/2)$

13- Determine whether these are valid arguments.

a) If x is a positive real number, then x^2 is a positive real number. Therefore, if a^2 is positive, where a is a real number, then a is a positive real number.
b) If $x^2 \neq 0$, where x is a real number, then $x \neq 0$. Let a be a real number with $a^2 \neq 0$; then $a \neq 0$.

14- Use resolution to show that the compound proposition $(p \vee q) \wedge (\neg p \vee q) \wedge (p \vee \neg q) \wedge (\neg p \vee \neg q)$ is not satisfiable.

15- Prove that if n is a perfect square, then n + 2 is not a perfect square.

16- Prove that there are no solutions in positive integers x and y to the equation $x^4 + 5y^2 = 14$.

17- Prove that these four statements about the integer n are equivalent: (i) n^2 is odd, (ii) 1 − n is even, (iii) n^3 is odd, (iv) $n^2 + 1$ is even.

18- Show that if r is an irrational number, there is a unique integer n such that the distance between r and n is less than 1/2.

19- Prove that there are no solutions in integers x and y to the equation $2x^2 + 5y^2 = 14$.

20- Prove or disprove that if you have an 8-gallon jug of water and two empty jugs with capacities of 5 gallons and 3 gallons, respectively, then you can measure 4 gallons by successively pouring some of or all of the water in a jug into another jug.